

Helix OTA “Session Resumption (canonical, always-current)

Field	Value
Revision	4
Last modified	2026-06-10T12:00:00Z
Status	active “ the §11.4.131 single canonical out-of-the-box session-resumption file
Standard path	docs/RESUMPTION.md (this file “ the fixed project-declared §11.4.131 entry point; do not move without a §11.4.66 operator decision)
Status summary	Point any fresh session at THIS file. It carries a SHORT one-line resume + a FULL block, the read-first handoff docs, exact live-state anchors, current PHASE + NEXT + terminal goal, and the binding constraints. Moment-valid for HEAD 4658125 (2026-06-10) + this RESUMPTION commit on top.

SHORT “ paste this first sentence into a fresh session

Read docs/research/main_specs/CONTINUATION.md first, run `git fetch --all --prune`, then continue the Helix OTA autonomous loop toward the next validated-and-published version tag “ current PHASE is “emulator-driven device testing (Tier-1 in progress)“; HEAD is 4658125 (+ this RESUMPTION commit) on main pushed to all 4 upstreams.

FULL “ detailed resumption block

0. Read FIRST (in order)

- docs/research/main_specs/CONTINUATION.md “ the live work-state handoff (DONE / NEXT / locked decisions). **This is the primary handoff doc.**
- .remember/remember.md “ present but currently empty; no extra state there.
- docs/design/EMULATED_DEVICE_TESTING.md “ the tiered emulator-testing plan (Tier-1/2/3) now in flight.
- Then run: `git fetch --all --prune` and reconcile HEAD. .@{u} before any edit (§11.4.37 fetch-before-edit).

1. Exact live-state anchors (moment-valid 2026-06-10)

- HEAD commit:** 4658125 “ fix(dashboard): DeviceStatus.health is an object “ fix Fleet-detail crash + cover (on main; this RESUMPTION commit lands on top).
- Wave 5 shipped (on main):** 2391cb6 full OTA lifecycle + multi-device fleet **PROVEN on podman** (single device 1.0.0†’1.1.0 with the complete download†’success telemetry

accepted; fleet 5/5 → 1.0.0 → 1.2.0; evidence docs/qa/20260610T11191*/ +
 ... 111928Z - fleet/); f5a3428 N-device scaling (50 concurrent, ~1580 ops/sec, 0
 err, -race); 4658125 dashboard Fleet-detail + a real product bug fix (DeviceStatus.health
 object). **The emulator now fully runs+tests the codebase end-to-end on podman →**
the prior → honest gap → is closed.

- **NEXT non-blocked candidates:** richer telemetry ingest fields (duration_ms/
 bytes_transferred → needs the device to send them); a rollback/recall lifecycle e2e on the
 emulator (drive a failure event → recall forward-fix); migrate the dashboard health-bug to
 a Fixed.md ATM entry. **BLOCKED:** ATM-003 Tier-2 (Linux+KVM Cuttlefish, design
 ready in docs/design/CUTTLEFISH_TIER2.md) + ATM-004 Tier-3 (RK3588 hardware).
- **Waves 3 → 4 also shipped (on main):** 3c57867 closed both protocol gaps
 (UpdateAvailable.deployment_id [ota-protocol → 7920842] + GET /deployments);
 8c0521d Tier-1 podman container e2e (PROVEN: control-plane container boots, ota-
 device-emu container runs the real round-trip → evidence docs/qa/20260610T105306Z/);
 5d4920e dashboard ArtifactUpload + populated-detail (Vitest 50, Playwright 20);
 submodule pins bumped → constitution ba0f702, ota-rollout-engine 7a90912, ota-
 artifact-validator 77c6b48, containers 845ad45. Tracker live at docs/Issues.md
 (ATM-003/004 Operator-blocked: Tier-2 Cuttlefish-on-Linux-KVM / Tier-3 RK3588
 hardware) + docs/Fixed.md.
- **Just-shipped this session (on main, pushed all 4 upstreams):**
 - 50ef5c6 → feat(api): per-device telemetry filters +
 group/members pagination (OpenAPI synced, redocly-clean).
 - b0b8ee2 → chore(dashboard): sync API client to new
 pagination/filter params (dashboard client lockstep).
 - 7dc3334 → feat(emulator): Tier-1 Go OTA device-emulator
 + resilience (server/internal/deviceemu + cmd/ota-device-
 emu; surfaced the telemetry deployment_id protocol gap).
 - fa571b8 → test(dashboard): comprehensive UI testing
 system (Vitest 43 + Playwright 17 + ally).
 - a839220 → test(qa): autonomous e2e (50/50) + security
 (39/39) + HelixQA bank.
 - containers submodule advanced to 845ad45 → feat(emulator):
 ota-device-emu runtime image + on-demand boot recipe
 (pushed to its github origin); parent gitlink bumped in this commit.
- **Branch:** main. All work merges to main; commit + push fan out to all upstreams.
- **Upstreams (4):** github (git@github.com:HelixDevelopment/
 helix_ota.git, also origin fetch), gitlab
 (git@gitlab.com:helixdevelopment1/helix_ota.git), gitflic
 (git@gitflic.ru:helixdevelopment/helix_ota.git), gitverse
 (git@gitverse.ru:helixdevelopment/helix_ota.git). origin push fans
 out to all four.
- **No in-flight background PIDs / detached pushes known at handoff time →** verify with
 git status + check qa-results/push_failures/ per \$11.4.88 before
 assuming clean.
- **Container runtime on this host:** podman (the pgx integration + dev stack boot via the
 containers submodule on podman). **No host QEMU / no AVD / no nested KVM on
 this Apple-Silicon host →** see binding constraints.

2. Current PHASE + immediate NEXT + terminal goal

- **PHASE:** Emulator-driven device testing → **Tier-1 in progress** (a podman container
 running a Go ota-device-emulator that speaks the real ota-protocol to the
 control plane). See docs/design/EMULATED_DEVICE_TESTING.md.

- **Immediate NEXT (priority order):**

1. Tier-1 emulator: stand up the podman ota-device-emulator exercising register â†’ update-check â†’ telemetry â†’ delta â†’ rollout â†’ recall against a live control plane, capturing real evidence under docs/qa/<run-id>/.
2. Remaining NEXT-wave items (hardware/ingest-gated): device-side TUF (gomobile-go-tuf/v2 â€” gated on real RK3588 .so/JNI measurement); device payload-apply integration (DeltaApplyDecision â†’ update_engine, needs a real device); row-4 richer telemetry fields (duration_ms/bytes_transferred â€” blocked on UNVERIFIED ingest).

- **Terminal goal (loop stop condition A, Â§11.4.126):** a new fully-validated-and-verified version (tag) created AND published across all owned submodules + main repo to all 4 upstreams.

3. Binding constraints (do not violate)

- **Anti-bluff covenant Â§11.4** â€” every PASS carries positive captured evidence; metadata-only / config-only / absence-of-error / grep-without-runtime PASS forbidden. Tests AND HelixQA Challenges bound equally.
- **Absolute no-force-push Â§11.4.113** â€” force-push (--force, --force-with-lease, +<ref>) is STRICTLY forbidden on every repo/submodule. Integrate via fetch â†’ merge-onto-latest-main â†’ fast-forward push.
- **podman-only on this host** â€” use the containers submodule (vasic-digital/containers, Â§11.4.76) for any containerized workload; never host-direct emulator/adb/qemu (Â§11.4.109 guard).
- **Tier-2 Android A/B is host-gated** â€” real update_engine A/B + AVB/dm-verity auto-rollback needs Cuttlefish on a Linux + nested-KVM box; the Apple-Silicon applehv host cannot run it. NOT structurally impossible (Â§11.4.112) â€” host/hardware-gated only.
- **Exact naming/versions:** Go module github.com/HelixDevelopment/helix_ota/server; submodules under submodules/; constitution submodule at constitution/. Toolchain: Go 1.26.2, Gradle 9.5 + Kotlin 2.3.20 (AGP 8.5.2 + plain kotlin.android builds; Kotlin MPP does NOT). No LaTeX/drawio/plantuml/graphviz/docker on host; PDF export needs a LaTeX engine (not installed).
- **Commit/push discipline Â§2.1 + Â§11.4.88** â€” commit + push to ALL 4 upstreams; pushes may run detached.

How a fresh session resumes from this file alone

Given only this fileâ€™s path, a new agent: reads Â§0 handoff docs, runs `git fetch --all`, confirms HEAD against Â§1, picks up the Â§2 PHASE/NEXT, and works under the Â§3 constraints toward the terminal goal â€” zero additional context required.